

Weight class prediction based on sparrow search algorithm optimised random forest model

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Abstract. In this paper, we improve the traditional random forest model by optimising the random forest algorithm based on sparrow search algorithm and compare the effectiveness of the two models for weight class prediction. Initial exploration of the data revealed that age, height, weight and BMI play an important role in weight class prediction. Correlation analyses showed a strong correlation between age and BMI and weight class. The experimental results show that the random forest model optimised based on the sparrow search algorithm achieves 100% in prediction accuracy, which improves the accuracy by 1.2% compared with the traditional random forest algorithm, and has better prediction effect. The significance of this paper is that a random forest algorithm optimised based on the sparrow search algorithm is proposed and experimentally demonstrated to have better performance in weight class prediction. This is of great significance in the fields of weight management, health assessment, and disease risk assessment. In addition, this study demonstrates the value of data analysis and machine learning methods in solving real-world problems. In conclusion, this paper provides new ideas for further improvement and application of machine learning algorithms, and provides references and lessons for researchers in related fields.

Keywords: Sparrow Search, Random Forest, BMI.

1. Introduction

BMI (Body Mass Index) is a commonly used body mass index to assess whether a person's weight is normal or not. It is calculated based on a person's height and weight. The significance of BMI analysis is to help people understand their weight status and provide reference to take appropriate health management measures [1].

Machine learning algorithms play an important role in predicting weight class. By using a large training dataset, machine learning algorithms can learn the relationship between different features and weight class and build predictive models. These features may include factors such as age, gender, height, and weight. By inputting data on these features, machine learning algorithms can make predictions based on existing patterns and regularities and give the corresponding weight class.

Using machine learning algorithms for prediction can help people assess their weight class more accurately. The traditional method of relying only on the BMI value to determine the weight class has some limitations, because it does not take into account other individual differences in factors. Machine

learning algorithms, on the other hand, can take multiple factors into account to provide more accurate and personalised predictions [2].

In conclusion, BMI analysis can help people understand their weight status, while machine learning algorithms have a more accurate and personalised role in predicting weight class. By using these tools and methods, people can better manage their health and take appropriate actions to improve their condition. In this paper, we propose a random forest algorithm optimised based on sparrow search algorithm based on the traditional random forest algorithm to improve the traditional random forest model and compare the effectiveness of the two models on weight class prediction.

2. Data sources

In this paper, we use the open source dataset on Kaggle to conduct experiments, the dataset is downloaded from (<https://www.kaggle.com/datasets/rukenmissonnier/age-weight-height-bmi-analysis>), the dataset has a total of 741 pieces of information, which records the age, height, weight, BMI and weight class of each measurer's age, height, weight, BMI and weight class [3], and some data are selected for presentation, and the results are shown in Table 1.

Table 1. Partial data.

Age	Height	Weight	Bmi	BmiClass
60	1.71	79.02	27.0236996	1
60	1.46	35.9	16.84180897	2
59	1.71	79.32	27.12629527	1
59	1.7	73.32	25.37024221	1
59	1.72	85.32	28.83991347	1
59	1.46	36	16.88872209	2
58	1.71	79.31	27.12287541	1
58	1.7	73.41	25.40138408	1
58	1.72	85.21	28.80273121	1
58	1.47	36	16.65972511	2
57	1.71	79.3	27.11945556	1

An initial exploration of the composition of the data was carried out by plotting violin plots for age, height, weight and BMI and observing the distribution of each variable, the results of which are shown in Figure 1.

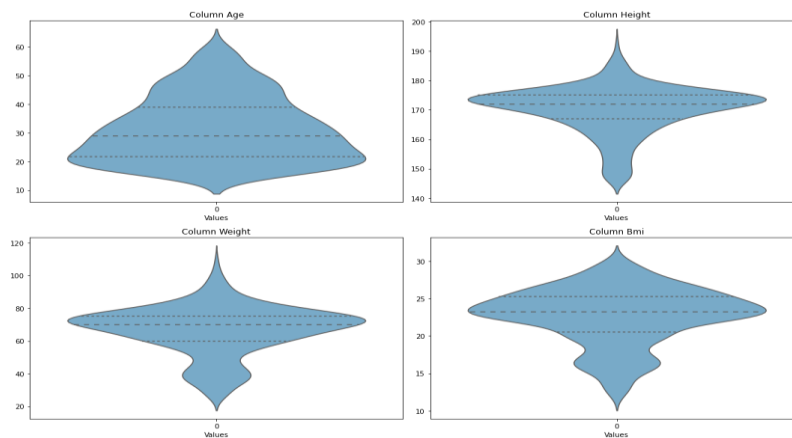


Figure 1. Violin diagram.

As can be seen from the violin chart, in terms of age distribution, there is a predominance of 20 years old spreading to both sides; in terms of height, there is a predominance of 175 spreading to both sides; and in terms of weight, there is a predominance of 70 kg spreading to both sides.

3. Relevance analysis

Correlation analysis is a statistical method used to measure the degree of association between two or more variables. Pearson's correlation coefficient is used to measure the linear relationship between two continuous variables and takes values ranging from -1 to 1, with positive values indicating a positive relationship, negative values indicating a negative relationship, and 0 indicating no linear relationship [4]. Correlation analysis measures and assesses the degree of association between variables by calculating correlation coefficients and performing statistical significance tests, and presenting the results through visualisation. We performed correlation analysis and drew correlation heat map for each tester's age, height, weight, BMI and weight class, and the results are shown in Fig. 2, and in order to facilitate the observation of the degree of correlation between each variable and weight class, we performed correlation ranking, and the results are shown in Fig. 3.

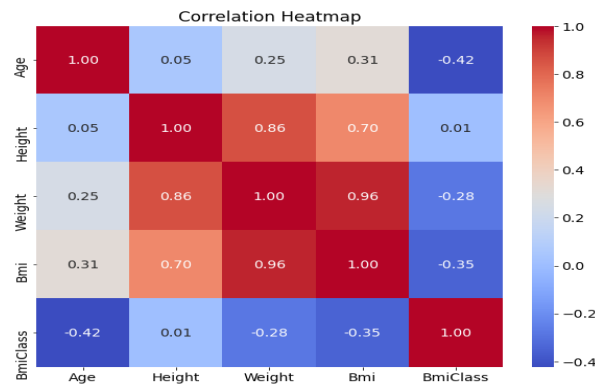


Figure 2. Correlation heat map.

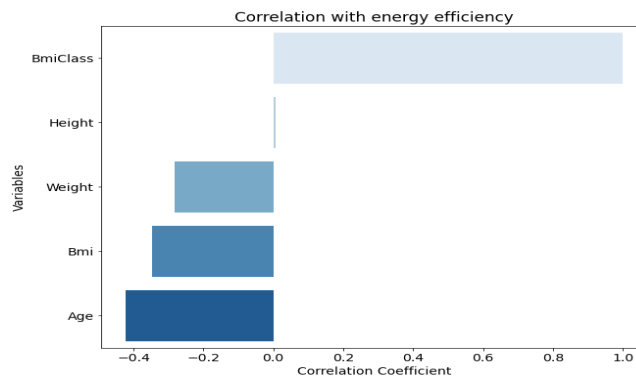


Figure 3. Correlation ranking.

From the correlation heat map, it can be seen that the variable with the greatest degree of correlation with the weight class is age, the absolute value of the correlation coefficient reaches 0.42, followed by BMI index, which reaches 0.31, the degree of correlation between body weight and the weight class is small, 0.25, and the degree of correlation between height and the weight class is the smallest, only 0.05.

4. Method

4.1. The random forest algorithm

The random forest algorithm is a powerful integrated learning method that performs classification and regression tasks by combining multiple decision trees. In constructing a random forest, a new training subset is first formed by randomly selecting a portion of samples from the original dataset (with putative sampling) [5]. Then a decision tree model is constructed for each training subset using a specific decision tree algorithm (e.g., CART). Finally, the final classification or regression results are obtained on the new input samples by making predictions on all the constructed decision trees and aggregating the predictions from each tree. Random forest algorithm has good generalisation ability and robustness and is suitable for dealing with high dimensional data and large scale datasets [6]. The schematic diagram of the random forest algorithm is shown in Figure 4.

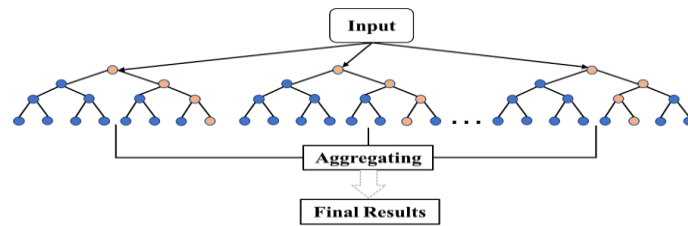


Figure 4. The schematic diagram of the random forest algorithm.

Random forest algorithm is effective because it uses the principle of ‘collective intelligence’. By combining multiple decision trees, each of which has its own bias and variance, but overall can correct each other's errors and achieve more accurate and stable prediction results [7]. In addition, Random Forest has advantages in feature selection and outlier handling. It can select the most important features by calculating feature importance, which reduces the risk of overfitting. Meanwhile, when dealing with outliers, it makes Random Forest more robust compared to a single decision tree due to methods such as random sampling and voting.

4.2. ResUnet

Sparrow search algorithm is an optimization algorithm based on the foraging behaviour of sparrows in nature, which can be applied to the optimization of random forest algorithm. The sparrow search algorithm performs optimisation by simulating the foraging behaviour of sparrows. In the random forest algorithm, each decision tree can be considered as a target to be foraged [8]. The sparrow search algorithm generates initial solutions (i.e., decision trees) by randomly generating them in the solution space and continuously improving the solutions (i.e., optimising the decision trees) by using local and global search strategies. In this way, after many iterations, the final solution obtained is an optimised random forest model.

Suppose the space consisting of N sparrows is $X=[x(1,i),x(2,i),...,x(N,i)]^T$, where $i=1,2,...,d$, and d is the dimension. Discoverer location update is described as shown in equation (1):

$$x_{i,j}^{t+1} = \begin{cases} x_{i,j}^t \cdot \exp(-\frac{i}{\alpha \cdot iter_{max}}) & \text{if } R_2 < ST \\ x_{i,j}^t + Q \cdot L & \text{if } R_2 \geq ST \end{cases} \quad (1)$$

The follower position update description is shown in equation (2):

$$x_{i,j}^{t+1} = \begin{cases} Q \cdot \exp(\frac{x_{worst}^t - x_{i,j}^t}{i^2}) & \text{if } i < \frac{N}{2} \\ x_p^{t+1} + |x_{i,j}^t - x_p^{t+1}| \cdot A^T (AA^T)^{-1} \cdot L & \text{otherwise} \end{cases} \quad (2)$$

In sparrow search algorithm, local search and global search are two important strategies. The local search strategy improves the current solution by adjusting the parameters and structure of the decision tree for better performance. Parameters such as threshold of split nodes, maximum depth or number of leaf nodes are adjusted in the decision tree [9]. The global search strategy, on the other hand, avoids falling into local optima and explores a wider solution space by introducing randomness and diversity. Methods such as random selection of feature subsets or training subsets are used in generating new solutions.

By applying the sparrow search algorithm to optimise the random forest, the accuracy and generalisation of the model can be improved and the risk of overfitting can be reduced. At the same time, the sparrow search algorithm can also achieve better results when dealing with large-scale datasets due to its better global exploration ability and convergence speed [10].

5. Result

This experiment was conducted using Matlab R2019b, in terms of data allocation, 60% of the data was used for the training set, 40% of the data was used for the test set, the initial population size was set to 6, the maximum number of iterations was set to 15, and the learning rate was set to 0.0001, and the Random Forest model and the optimised Random Forest algorithm of the sparrow search algorithm were used with the same environment configurations and parameter settings, and the two experiments are run in the same environment.

The confusion matrices of the results predicted by the test set of the two models are output, the confusion matrix predicted by the random forest model is shown in Figure 5, and the confusion matrix of the sparrow search optimised random forest algorithm is shown in Figure 6. The prediction effectiveness of the models was evaluated using precision, accuracy, recall and F1 score and the results are shown in Table 2.

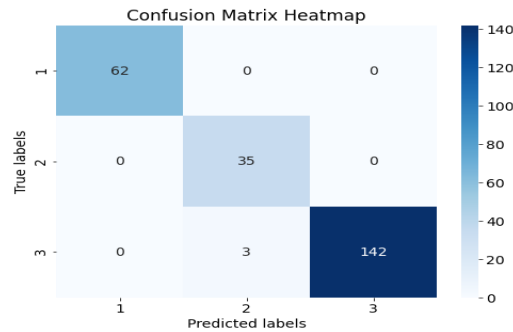


Figure 5. Confusion matrix.

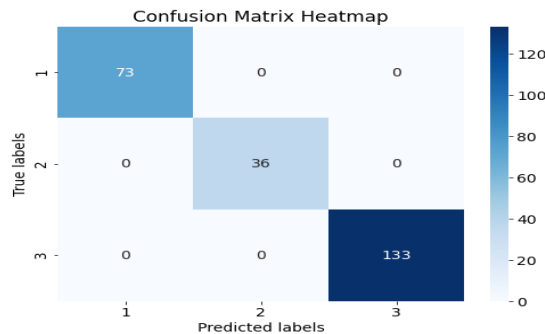


Figure 6. Confusion matrix.

Table 2. Indicators for model evaluation.

Model	Accuracy	Recall	Precision	F1 score
Random forest	0.988	0.988	0.989	0.988
Random Forests for Sparrow Search Optimisation	1	1	1	1

From the results of the test set, it can be seen that the random forest algorithm has 98.8% accuracy, 98.8% recall, 98.9% recall and F1 score of 1 in weight class prediction, while the random forest model optimised based on the sparrow search algorithm achieves 100% in prediction accuracy, which is 1.2% more accurate than the traditional random forest algorithm, and the improved model has a better prediction effect.

6. Conclusion

Machine learning algorithms play an important role in predicting weight class. In this paper, a random forest algorithm optimised based on the sparrow search algorithm is proposed to improve the traditional random forest model, and the effects of the two models on weight class prediction are compared.

Firstly, the distributions of age, height, weight and BMI are demonstrated by initial exploration of the data using violin plots. Observation of the violin plots reveals that in the age distribution, it is mainly concentrated around 20 years old and spreads to both sides; the height distribution is dominated by 175 cm and shows a trend of spreading to both sides; and the weight distribution is dominated by 70 kg and shows a trend of spreading to both sides.

Then, correlation analysis was conducted for age, height, weight, BMI and weight class of each tester, and correlation heat map was drawn. The correlation heat map shows that the most correlated variable with weight class is age, with a correlation coefficient of 0.42; followed by BMI, with a correlation coefficient of 0.31; whereas the correlation between weight and weight class is smaller, at 0.25; and the correlation between height and weight class is the smallest, at 0.05.

Then, this paper compares the effectiveness of the traditional random forest model and the optimised random forest algorithm based on the sparrow search algorithm on weight class prediction using the same environment configuration and parameter settings. The test set results show that the traditional random forest algorithm achieves 98.8%, 98.8% and 98.9% in accuracy, recall and F1 score respectively, while the optimised random forest model based on the sparrow search algorithm achieves 100% accuracy. This means that the improved model improves the accuracy by 1.2% compared to the traditional method and performs better in prediction.

In summary, this paper achieved significant improvement in the weight class prediction task by introducing the sparrow search algorithm to optimise the traditional random forest model. The method is able to predict an individual's weight class more accurately and has high accuracy and recall. This study provides useful ideas and methods to further enhance the application of machine learning algorithms in predicting weight class.

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